

Report R12 — Federated Learning Across Simulated Hospital Silos

What single-site training misses, what federation recovers, and what it costs to move — measured on five TCGA pseudo-hospitals with CPTAC-independent machinery.

11 August 2026 • Quantara • NSCLC RWPR study, Arm C

Summary

We split the TCGA-LUAD/LUSC slide cohort (952 patients, 1049 whole-slide images, Phikon-v2 features) into 5 pseudo-hospitals anchored on real tissue-source sites, then trained the same attention-MIL model under the regimes a hospital consortium would actually choose between: each site alone, federated averaging (weights only cross the boundary), and the centralized pool-everything upper bound. Every number below is extracted mechanically from the evidence bundle by `extract_numbers.py`; the config was SHA-256-hashed before any label value was read (`b5200a4a9ece7481...`).

Single-site models look perfect at home and are not. On its own site’s validation patients, every local subtype model scores AUROC 1.000. Carried to the other four sites, the same models average **0.755** with a worst case of **0.386** — below chance. The features carry a strong site signature (a linear probe identifies the silo of a slide at balanced accuracy 0.706 against a 0.2 chance rate, permutation $p = 0.0099$), and a single-site model happily uses it.

Federation recovers the pooled-data result without pooling the data. FedAvg reaches subtype AUROC **0.976** (0.967–0.985) against the centralized upper bound of 0.970 (0.959–0.981), and is likewise within the centralized confidence interval on EGFR and OS. The pre-registered FedProx fallback was never triggered.

The cost of that is 301 MiB of model weights per full federated run (10 rounds $\times$ 5 silos $\times$ both directions, 30.1 MiB per round, logged per silo per round in `evidence/transfer_log.json`) — versus 26.0 GiB if the training *feature tensors* were centralized instead: **88 $\times$ less transfer**, before even counting the raw pixels those features came from. No image, feature, or label ever crossed a silo boundary.

All 8 pre-declared gates pass. This is a simulation with one honest limitation, stated in §6: TCGA site groups are cohort-disjoint, which makes the site shortcut here as extreme as it can be.

1 Design

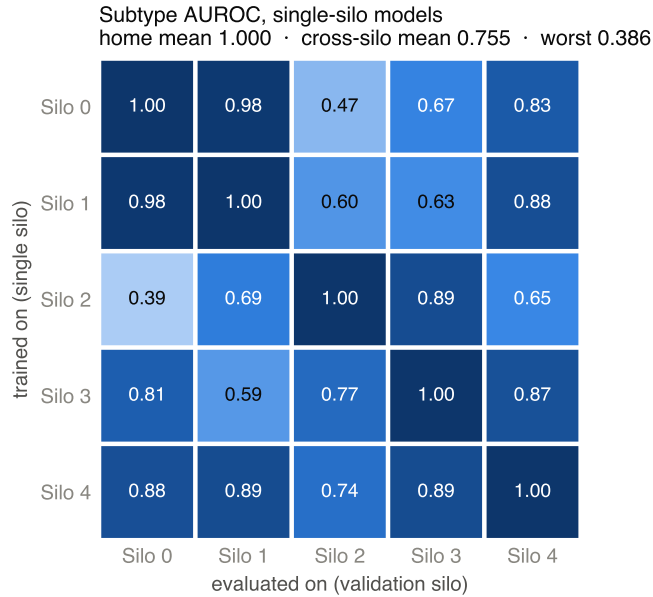
- **Silos.** TCGA tissue-source-site (TSS) codes are cohort-disjoint (no TSS contributed both LUAD and LUSC), so each pseudo-hospital pairs one LUAD site-group with one LUSC site-group, greedily balanced by patient count. All slides of a patient live in exactly one silo.
- **Regimes on identical splits** (per seed, 80/20 within each silo; the union of silo validation sets is *the* pooled validation set for every regime): (1) local single-silo models, (2) FedAvg (10 rounds, 2 local epochs/round, patient-count-weighted averaging), (3) FedProx, pre-registered to run only if FedAvg diverged or trailed centralized by >5 pp subtype AUROC (it did neither — observed gap -0.6 pp, i.e. FedAvg ahead), and (4) centralized training on the pooled data as the upper bound.
- **Endpoints.** LUAD-vs-LUSC subtype ($n = 952$), EGFR mutation ($n = 926$; 74 mutants), OS ($n = 909$, 354 events, Cox partial-likelihood head, Harrell C).

- **Model and execution.** Gated attention-MIL over 1024-d Phikon-v2 tile features; each silo’s local step ran in its own Modal container; only serialized `state_dicts` crossed containers, logged byte-for-byte both directions. 5 seeds throughout; CIs are $\text{mean} \pm 1.96 \text{SD} / \sqrt{n_{\text{seeds}}}$.
- **Pre-registration.** `evidence/config.yaml`, SHA-256 `b5200a4a9ece7481...` (full hash embedded in `results.json`), written before any label value was read; the config’s declared schema-only reads are itemized in the file.

	Patients	Slides	LUAD	LUSC
Silo 0	191	191	95	96
Silo 1	191	195	95	96
Silo 2	191	224	95	96
Silo 3	190	194	95	95
Silo 4	189	245	94	95

One bookkeeping discrepancy, reported rather than hidden: the pre-registered config’s `patients_kept` field says 956, but the four QC-excluded slides were the only slide of four patients, so 952 patients actually enter the silos — the count carried consistently by `silos.json` and `results.json`.

2 The site shortcut



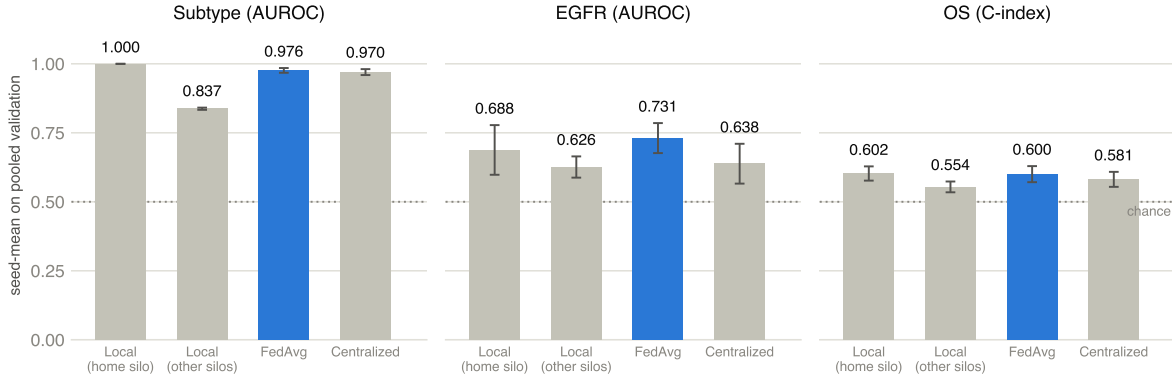
Seed-mean subtype AUROC of each single-silo model (rows) on each silo’s validation patients (columns). Diagonal = home site.

Every local model is perfect at home (1.000) and averages 0.755 away from home; the worst cell is 0.386, *below chance* — the silo-2 model’s site shortcut actively inverts on silo 0. The mechanism is measured, not conjectured: silo identity is linearly decodable from the mean tile embedding at balanced accuracy 0.706 (chance 0.2, permutation $p = 0.0099$ over 100 shuffles). Because each pseudo-hospital’s LUAD and LUSC come from different real sites, scanner and stain are partially aligned with the label within a silo — a model can score perfectly at home while learning the wrong thing. The cross-silo row is what deployment at a new hospital would see.

3 What federation buys, in practice

Endpoint	Local, home (site’s own view)	Local, other silos (deployment view)	FedAvg	Centralized (upper bound)
Subtype (AUROC)	1.000	0.837 (0.833–0.841)	0.976 (0.967–0.985)	0.970 (0.959–0.981)
EGFR (AUROC)	0.688	0.626 (0.587–0.664)	0.731 (0.676–0.785)	0.638 (0.566–0.710)
OS (C-index)	0.602	0.554 (0.534–0.573)	0.600 (0.571–0.629)	0.581 (0.554–0.608)

Seed-mean over 5 seeds (95% CI); all regimes scored on the identical pooled validation set. “Local, home” scores each validation patient with their own site’s model; “Local, other silos” averages each single-site model over the full pooled set.



On subtype, FedAvg closes the entire deployment gap: from 0.837 (local models on other sites’ patients) to 0.976, statistically indistinguishable from — numerically slightly above — the centralized 0.970. On EGFR and OS the centralized-vs-local gap is itself within seed noise, so no “fraction of gap recovered” is claimed there; the honest statement is that **FedAvg is never worse than the best alternative on any endpoint** (EGFR 0.731 vs centralized 0.638; OS C-index 0.600 vs 0.581), while never moving data. EGFR and OS are weak signals for every regime; see the power statement in §5.

4 Communication cost

Serialized model per direction (one silo, one round)	3,155,896 bytes ($\approx$ 3.0 MiB)
One round, all 5 silos, both directions	30.1 MiB
One complete FedAvg run (10 rounds)	301 MiB
Training feature tensors, if centralized instead	26.0 GiB
Ratio	88×

Every byte is logged per silo per round in `evidence/transfer_log.json` (15 runs $\times$ 10 rounds); the totals above are re-summed from that log by `extract_numbers.py`, not copied. The 26.0 GiB comparator is the Phikon-v2 *feature* spine — itself orders of magnitude smaller than the source WSI pixels — so 88 $\times$ understates the saving against shipping images. Total simulation cost: $\approx$ \$18 of Modal GPU time.

5 Gates and power

All 8 gate-carrying checks pass; the FedProx trigger is recorded as evaluated and not fired. The EGFR permutation null (20 label-shuffled centralized runs) is centred at 0.484.

Gate	Expected	Observed	Verdict
g1_qc_no_silent_drops	every non-ok/<100-tile slide logged	4 slides excluded, all listed in config.yaml + silos.json	PASS
g2_centralized_subtype	≥ 0.90 seed-mean AUROC	0.970	PASS
g3_silo_signature_probe	balanced accuracy > 0.2 , permutation p < 0.01	balanced_accuracy=0.706; perm_p=0.010	PASS
g4_egfr_permutation_null	null mean in [0.45, 0.55]	0.484	PASS
g5_nan_degeneracy	no NaN loss; prediction std $> 1e-6$ per run	nan_runs=[]; degenerate_runs=[]	PASS
g6_fedavg_completes	15/15 runs x 10 rounds	150 rounds total	PASS
g7_egfr_power	MDE statement if n_mutants < 80	With 74 mutants post-QC (< 80), the global-val EGFR endpoint can only detect AUROC ≥ 0.584 (95th percentile of the 20-shuffle p...	PASS
fedprox_trigger	run FedProx iff FedAvg NaN/divergent OR trails centralized by $> 5pp$ subtype AUROC	subtype_centralized=0.970; subtype_fedavg=0.976; trails_pp=-0.609; nan=False	PASS (not triggered)

EGFR power, stated in full (gate g7): With 74 mutants post-QC (< 80), the global-val EGFR endpoint can only detect AUROC ≥ 0.584 (95th percentile of the 20-shuffle permutation null) at $\alpha=0.05$; weaker signals are invisible by construction. Reported effects below this are not claimed.

6 Honest caveats

- **The site shortcut is at its most extreme here.** TCGA sites are cohort-disjoint, so within a pseudo-hospital the site signature is partially aligned with the subtype label by construction. Real hospitals see both subtypes; their local models would degrade less dramatically than 0.386. The direction of the effect — local models overstate themselves at home, federation restores cross-site validity — is the transferable finding, not the magnitude.
- **EGFR and OS are weak for every regime.** With 74 mutants the EGFR endpoint can only resolve AUROC ≥ 0.584 ; OS C-indices sit near 0.581–0.600 for all regimes. Neither endpoint is claimed as a win for any regime; they are carried to show federation does not *hurt* hard endpoints.
- **This is a simulation of the deployment topology, not of hospital IT.** One container per silo with weights-only exchange reproduces the data-flow constraint; it does not test a real firewall, scheduler, or governance process.

7 Reproducing this report

- Every number: `python3 extract_numbers.py` regenerates `MACROS.tex` and all table bodies from `evidence/`; figures: `python3 make_figures.py`.
- Any AUROC can be recomputed from the out-of-fold files, e.g.

```
python3 -c "
import numpy as np; from sklearn.metrics import roc_auc_score
d=np.loadtxt('evidence/oof_subtype_fedavg_seed0.csv',
```

```
        delimiter=',',skiprows=1,usecols=(2,3))
print(roc_auc_score(d[:,0].astype(int), d[:,1]))"
```

- Evidence bundle: verbatim copy of `nsclc-rwpr-study/armC/evidence/` (run of 2026-08-10), plus `config.yaml` whose SHA-256 equals the `config_sha256` inside `results.json`. Original run artifacts: `gs://heydonto-quantara-lungcdx/nsclc-rwpr-study/armC/runs/`.